ivdtools

Statistical tools for the evaluation of *in vitro* diagnostic reagents

`v0.1.1` `R ≥ 4.1.0` `CRAN` `github.com/hiox-tech/ivdtools` `MIT`

GET STARTED

`install.packages("ivdtools")`	from CRAN
`library(ivdtools)`	load

`mcr()` → `analyse()` → `plot()` / `summary()`

`ivd_mcr_example` `ivd_roc_example` `ivd_qualitative_example` `ivd_bottle_example`

workflow: build an S3 object, one analysis per call — each returns an **updated object; reassign**. Results don't auto-judge — interpret against pre-specified limits.

INSPECT & PREPARE

`outliers_test(df, "v", method = "grubbs")`	Grubbs · ESD · Dixon · IQR
`normal_test(df, "v")`	Shapiro / AD / Lilliefors / CVM
`plot(nt, type = "qq")`	Q-Q plot or histogram
`replicate_to_mean(d, "x", "y", weights = "1/sd^2")`	mean ± SD per sample
`continuous_to_binary(d, "marker", cutoff = 10)`	make a 0/1 reference

normal_test auto-selects by n; outliers_test takes `alpha`, `r` (ESD), `type` (Dixon).

METHOD COMPARISON — EP09

`mcr(df, id, candidate, reference)`	create object
`describe(obj)`	descriptive stats
`correlation(obj, method = "pearson")`	Pearson / Spearman / Kendall
`regression(obj, method = "deming")`	OLS · WLS · Deming · wDeming · PB
`outlier(obj, method = "grubbs")`	on difference or % diff
`bland_altman(obj, type = "difference")`	LoA: diff / ratio / percent
`bias(obj, mdl = c(15, 25))`	bias at decision levels
`predict(obj, newdata, inverse = TRUE)`	score new samples
`plot(obj, type = "regression")`	scatter · reg · BA · bias
`summary(obj)`	consolidated report

order: `regression()` first, then `bias()` / `predict()`. Set `lambda` / `weights` for wDeming.

SAMPLE SIZE

`sample_size_bland_altman(power = .8, sd = 4, delta = 2)`	n for LoA (or power given n)
`sample_size_proportion(p0 = .2, p1 = .35, power = .8)`	n for a proportion diff
`sample_size_proportion_ci(p = .3, d = .05)`	n for CI half-width

CI methods: wilson · wald · wald-cc · agresti-coull · jeffreys · clopper-pearson.

PRECISION — EP05

`precision(df, y ~ day/run, by = "sample")`	variance components
`variance(obj)` `vc(obj)`	SD / CV table
`ci(obj)`	Satterthwaite CI for SD / CV
`normal(obj)`	within-group normality
`outlier(obj, method = "grubbs")`	Grubbs or IQR
`profile(obj, model.no = 1:10)`	Sadler precision profile
`plot(obj, type = "var")`	dot · hist · var · qq · profile
`summary(obj)`	per-step status
`list_sadler()`	the 10 candidate models

formula: nested factors e.g. `y ~ lot/calibration/day/run`. `profile()` needs `variance()` first.

LINEARITY & CURVE FITTING — EP06

`list_equation(category = "Response")`	registry E01–E12 + Sadler
`fit_equation("4PLC", d, x = "Con", y = "OD")`	by ID · name · category
`fit_equation("y ~ a*(1-exp(-b*x))", d)`	custom formula
`compare_equation(d, eqs = c("4PLC", "linear"))`	rank candidate curves
`coef(fit)` `predict(fit, inverse = TRUE)`	parameters · estimate x from y
`residuals(fit)` `plot(fit, interval = "confidence")`	residuals · curve + bands

weights: `"equal"` · `"1/y"` · `"1/y^2"` · `"1/x"` · `"1/x^2"`; bounds via `lower` / `upper` / `constraints`.

BOTTLE-TO-BOTTLE — EP15

`bottle_anova(df, value ~ bottle)`	one-way ANOVA
`bottle_anova(df, value ~ batch/vial)`	nested design
`tukey(obj, term = "bottle")`	Tukey HSD pairwise

Variance decomposition + F-tests between bottles; `conf.level` sets the CIs.

QUALITATIVE AGREEMENT — EP12

`raw_to_table(df, "new", "gold", positive = "positive")`	2×2 from raw pairs
`counts_to_table(tp, fp, tn, fn)`	2×2 from counts
`describe(tbl)`	counts · rates · marginals
`diagnostics(tbl)`	Se · Sp · PPV · NPV · LR · Youden
`kappa(tbl)`	Cohen's κ + CI
`mcnemar(tbl)`	paired disagreement test
`summary(tbl)`	combined table report

REFERENCE INTERVAL — EP28

`reference_interval(df, "val", method = "all")`	RI + CI for limits
`plot(ri)`	histogram with limits

`interval = 0.95` coverage · `ci = 0.95` limits CI · id duplicates · `method = "all"` runs all three.

ANALYTICAL SENSITIVITY — EP17

`lob_lod_loq(sum, blank = "Blank")`	LoB · LoD · LoQ
`plot(s)`	precision profile

input: summarized `mean` / `sd` / `n` per sample — defaults match `replicate_to_mean()`. `target_cv = 0.2` sets the LoQ precision target.

ROC ANALYSIS — EP24

`roc(df, cols = c("m1", "m2"), reference = "ref")`	1+ markers vs binary ref
`continuous_to_binary(d, "ref", cutoff)`	binarize a continuous ref
`auc(obj)`	per-marker AUC ± CI
`cutoff(obj)`	optimal cutoff + metrics
`mlr(obj, name = "fit")`	logistic regression
`predict(obj, newdata)`	score new data from MLR
`plot(obj, cutpoint = "Youden")`	curves · overlay MLR
`describe(obj)` `summary(obj)`	stats · combined report

reference: 0/1, 2-level factor, or 2-value character; set `positive`. Direction is auto-detected.

STABILITY STUDIES — EP25

`stability_bias(df, condition, time, value)`	bias vs reference condition
`stability_regression(df, time, value, condition)`	trend fit · self / compare
`stability_time(sr, limit = 10)`	estimate shelf life
`stability_plan(allowable_drift = 10, variability = 2)`	design time points
`arrhenius(df, temp, time, val, target, limit)`	accelerated stability
`mkt(df, temp_cols, time)`	mean kinetic temperature
`plot(obj)`	each step has a plot method

settings: `bias_type = "relative"/"absolute"`, `time_unit`, `ea` (activation energy, kJ/mol).

QUALITY CONTROL — Westgard

`list_westgard()`	rule names + meanings
`qc_chart(df, "value", group = "level")`	multi-rule Levey–Jennings
`plot(qc)`	chart with violations
`youden_plot(df, "s1", "s2")`	two-sample Youden plot

default rules: `"1-2s,1-3s,2-2s,R-4s,4-1s,10x"`. Pass `mean` / `sd` for target values instead of estimating them.

ANALYSIS WORKFLOW

Load data → Inspect → Analyse → Visualise → Report

csv / xlsx / rds · outliers & normality · module fn · `plot()` · `summary()` / HTML

AI-ASSISTED ANALYSIS

The `ivdtools-analysis` skill turns plain-language IVD evaluation requests into reproducible, CLSI-aligned analyses.

- R code · result tables · publication-quality figures
- Method comparison · precision · ROC · RI · stability · sensitivity
- Reproducible Chinese HTML reports

CLSI MODULE MAP

`EP05 Precision` `EP06 Linearity` `EP09 Comparison` `EP12 Qualitative` `EP15 Bottle-to-bottle` `EP17 Sensitivity` `EP24 ROC` `EP25 Stability` `EP28 Reference interval`

Results are interpreted against pre-specified acceptance limits — the package does not auto-judge pass/fail.